LECTURE 4: BUILDING E-COMMERCE PRESENCE

- Building an E-commerce Presence: Web Sites, Mobile Sites, and Apps

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Lecture 4

- Objectives

- Discuss the questions you must ask and answer, and the steps you should take, in developing an e-commerce presence.
- Explain the process that should be followed in building an e-commerce presence.
- Describe the major issues surrounding the decision to outsource site development and/or hosting.

Lecture 4: objectives

- Identify and understand the major considerations in choosing Web server and e-commerce merchant server software.
- Explain the issues involved in choosing the most appropriate hardware for an e-commerce site.
- Identify additional tools that can improve Web site performance.
- Explain the important considerations involved in developing a mobile Web site and building mobile applications.

Imagine Your E-commerce Presence

- **What's the idea?**
  - Vision
  - Mission statement
  - Target audience
  - Intended market space
  - Strategic analysis
  - Internet marketing matrix
  - Development timeline and preliminary budget

Imagine Your E-commerce Presence (cont.)

- **Where's the money?**
  - Business model(s):
    - Portal, e-tailer, content provider, transaction broker, market creator, service provider, community provider
  - Revenue model(s):
    - Advertising, subscriptions, transaction fees, sales, and affiliate revenue

Imagine Your E-commerce Presence (cont.)

- **Who and where is the target audience?**
  - Describing your audience
    - Demographics
      - Age, gender, income, location
    - Behavior patterns (lifestyle)
    - Consumption patterns (purchasing habits)
    - Digital usage patterns
    - Content creation patterns (blogs, Facebook)
    - Buyer personas:-Do you know who your business's buyer personas are? And exactly how much do you know about them?
      - Buyer personas (sometimes referred to as marketing personas) are fictional, is a semi-fictional representation of your ideal customer based on market research and real data about your existing customers.
      - When creating your buyer persona(s), consider including customer demographics, behavior patterns, motivations, and goals. The more detailed you are, the better.

Imagine Your E-commerce Presence (cont.)

- A buyer persona is a semi-fictional representation of your ideal customer based on market research and real data about your existing customers.
- When creating your buyer persona(s), consider including customer demographics, behavior patterns, motivations, and goals. The more detailed you are, the better.

Imagine Your E-commerce Presence (cont.)

- **Characterize the marketplace**
  - Demographics:-
    Socioeconomic characteristics of a population expressed statistically, such as age, sex, education level, income level, marital status, occupation, religion, birth rate, death rate, average size of a family, average age at marriage. A census is a collection of the demographic factors associated with every member of a population
  - Size, growth, changes
  - Structure
    - Competitors
    - Suppliers
    - Substitute products
- **Where is the content coming from?**
  - Static or dynamic?

Imagine Your E-commerce Presence (cont.)

- **Know yourself—SWOT analysis**
- **Develop an e-commerce presence map**
- **Develop a timeline: Milestones**
- **How much will this cost?**
  - Simple Web sites: up to ksh500,000
  - Small Web start-up: ksh2,500,000 to ksh5,000,000
  - Large corporate site: ksh10,000,000+ to billions

SWOT Analysis



Figure 4.1, page 189

E-commerce Presence Map

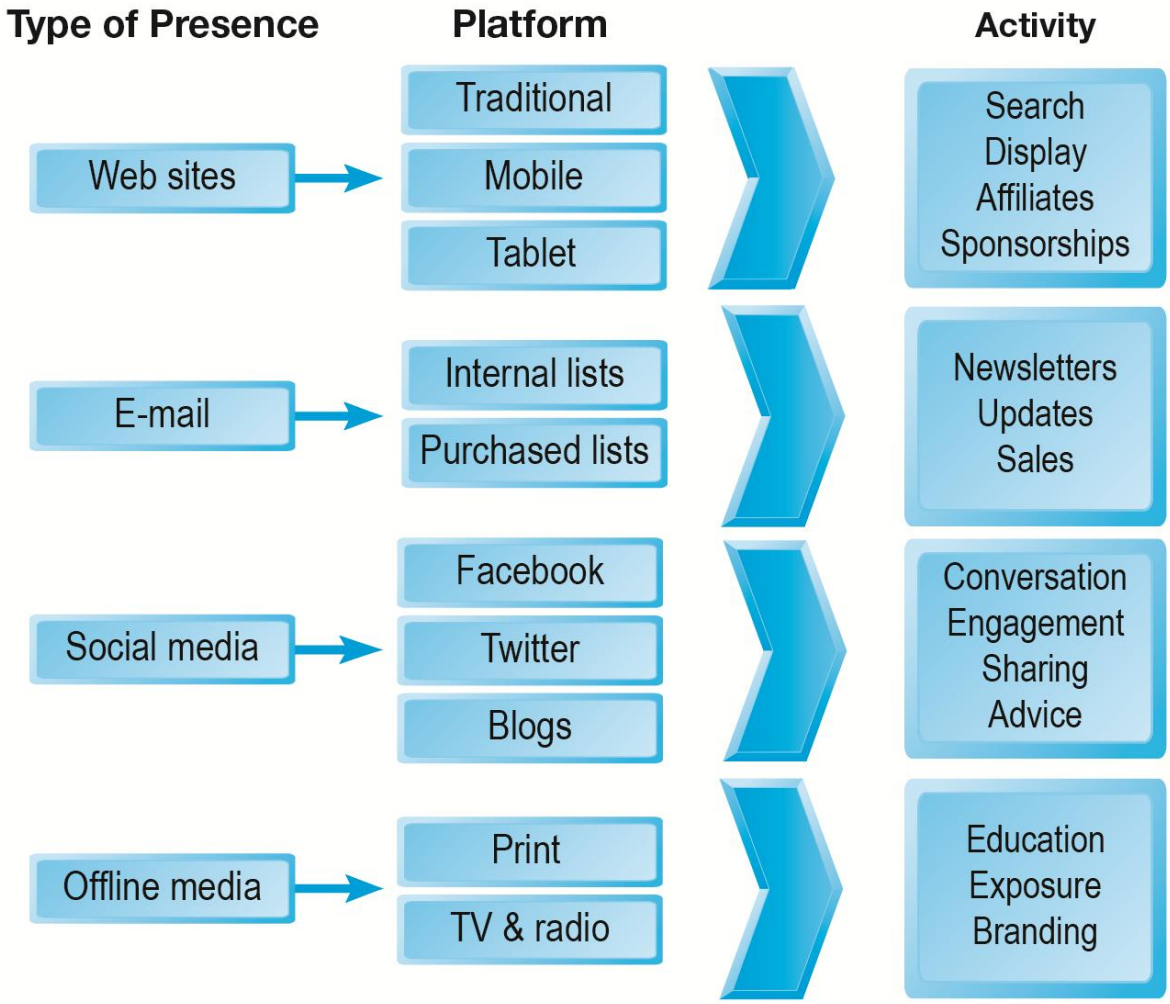


Figure 4.2, page 190

Building an E-commerce Site: A Systematic Approach

- **Most important management challenges:**
  - Developing a clear understanding of business objectives
  - Knowing how to choose the right technology to achieve those objectives

Pieces of the Site-Building Puzzle

- **Main areas where you will need to make decisions:**
  - Human resources and organizational capabilities
    - Creating team with skill set needed to build and manage a successful site
  - Hardware/software
  - Telecommunications
  - Site design

The Systems Development Life Cycle

- **Methodology for understanding business objectives of a system and designing an appropriate solution**
- **Five major steps:**
  - Systems analysis/planning
  - Systems design
  - Building the system
  - Testing
  - Implementation

Web Site Systems Development Life Cycle

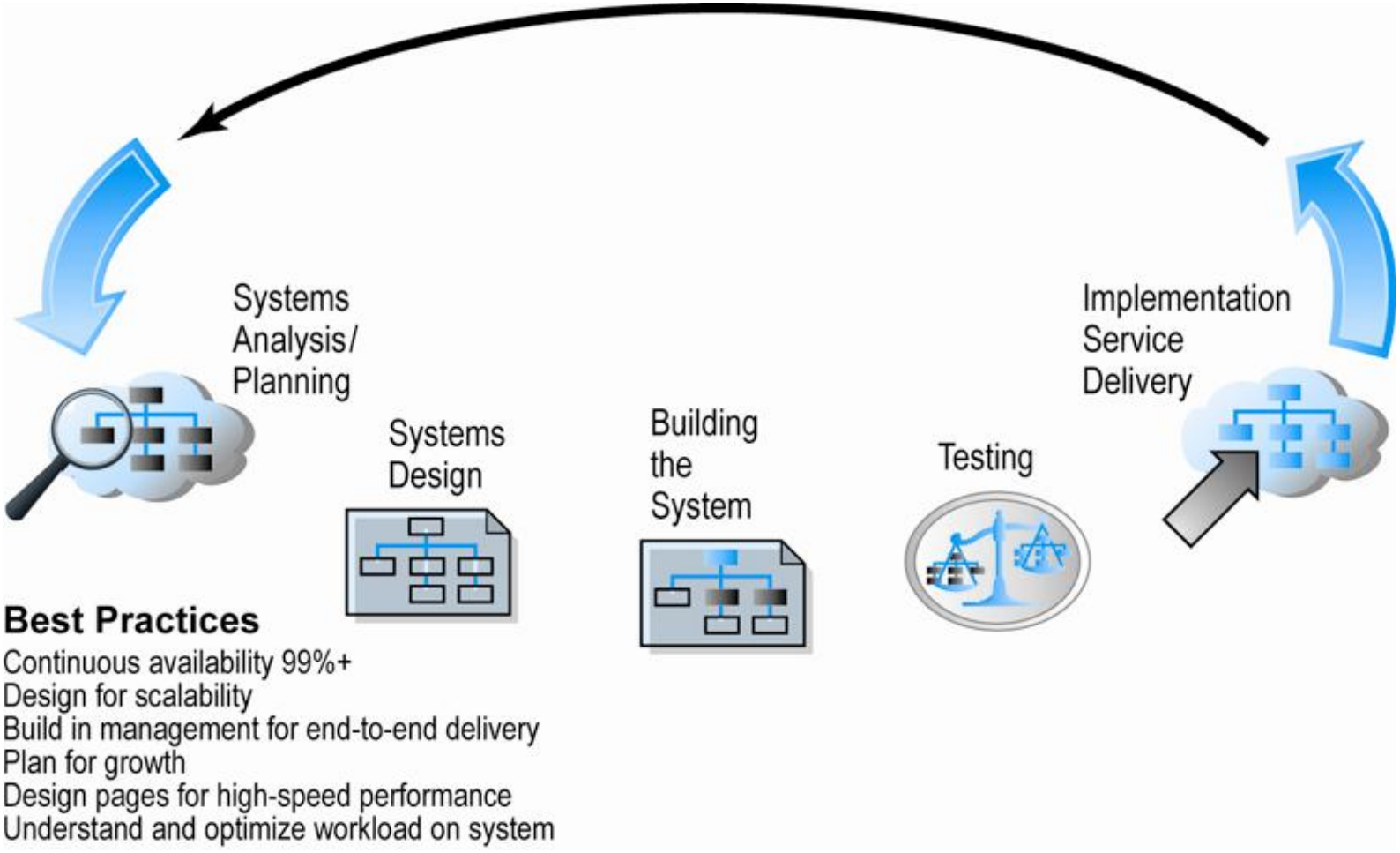


Figure 4.5, Page 194

System Analysis/Planning

- **Business objectives:**
  - List of capabilities you want your site to have
- **System functionalities:**
  - List of information system capabilities needed to achieve business objectives
- **Information requirements:**
  - Information elements that system must produce in order to achieve business objectives

TABLE 4.2 SYSTEM ANALYSIS: BUSINESS OBJECTIVES, SYSTEM FUNCTIONALITIES, AND INFORMATION REQUIREMENTS FOR A TYPICAL E-COMMERCE SITE

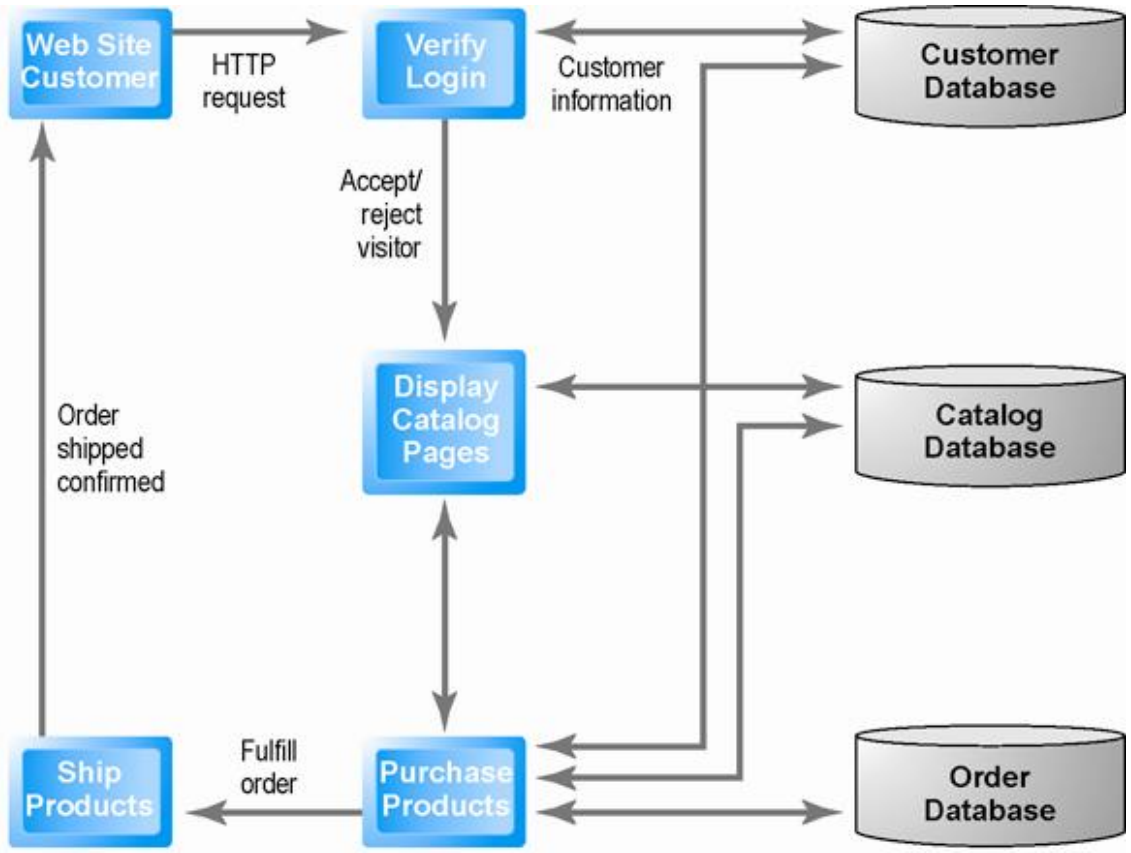
BUSINESS OBJECTIVE	SYSTEM FUNCTIONALITY	INFORMATION REQUIREMENTS
Display goods	Digital catalog	Dynamic text and graphics catalog
Provide product information (content)	Product database	Product description, stocking numbers, inventory levels
Personalize/customize product	Customer on-site tracking	Site log for every customer visit; data mining capability to identify common customer paths and appropriate responses
Engage customers in conversations	On-site blog	Software with blogging and community response functionality
Execute a transaction	Shopping cart/payment system	Secure credit card clearing; multiple payment options
Accumulate customer information	Customer database	Name, address, phone, and e-mail for all customers; online customer registration
Provide after-sale customer support	Sales database	Customer ID, product, date, payment, shipment date
Coordinate marketing/advertising	Ad server, e-mail server, e-mail, campaign manager, ad banner manager	Site behavior log of prospects and customers linked to e-mail and banner ad campaigns
Understand marketing effectiveness	Site tracking and reporting system	Number of unique visitors, pages visited, products purchased, identified by marketing campaign
Provide production and supplier links	Inventory management system	Product and inventory levels, supplier ID and contact, order quantity data by product

Table 4.2, page 195

Systems Design: Hardware and Software Platforms

- **System design specification:**
  - Description of main components of a system and their relationship to one another
- **Two components of system design:**
  - Logical design
    - Data flow diagrams, processing functions, databases
  - Physical design
    - Specifies actual physical, software components, models, and so on

Logical Design for a Simple Web Site



(a) Simple Data Flow Diagram

This data flow diagram describes the flow of information requests and responses for a sample Web site

Figure 4.6 (a), Page 197

Physical Design for a Simple Web Site

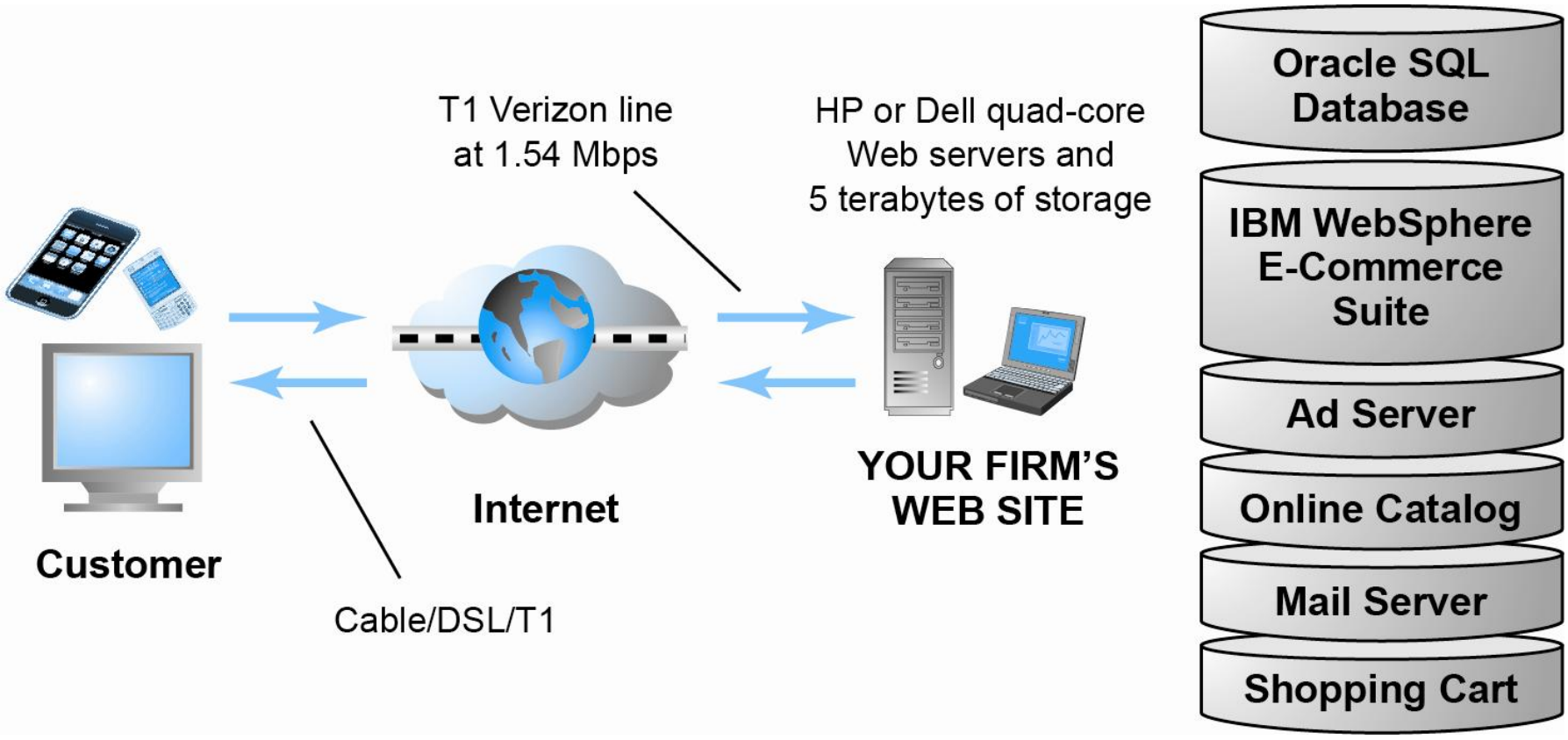


Figure 4.6 (b), Page 197

Build/Host Your Own vs. Outsourcing

- **Outsourcing: Hiring vendors to provide services involved in building site**
- **Build own vs. outsourcing:**
  - Build your own requires team with diverse skill set; choice of software tools; both risks and possible benefits
- **Host own vs. outsourcing**
  - **Hosting:** Hosting company responsible for ensuring site is accessible 24/7, for monthly fee
  - **Co-location:** Firm purchases or leases Web server (with control over its operation), but server is located at vendor's facility

Choices in Building and Hosting

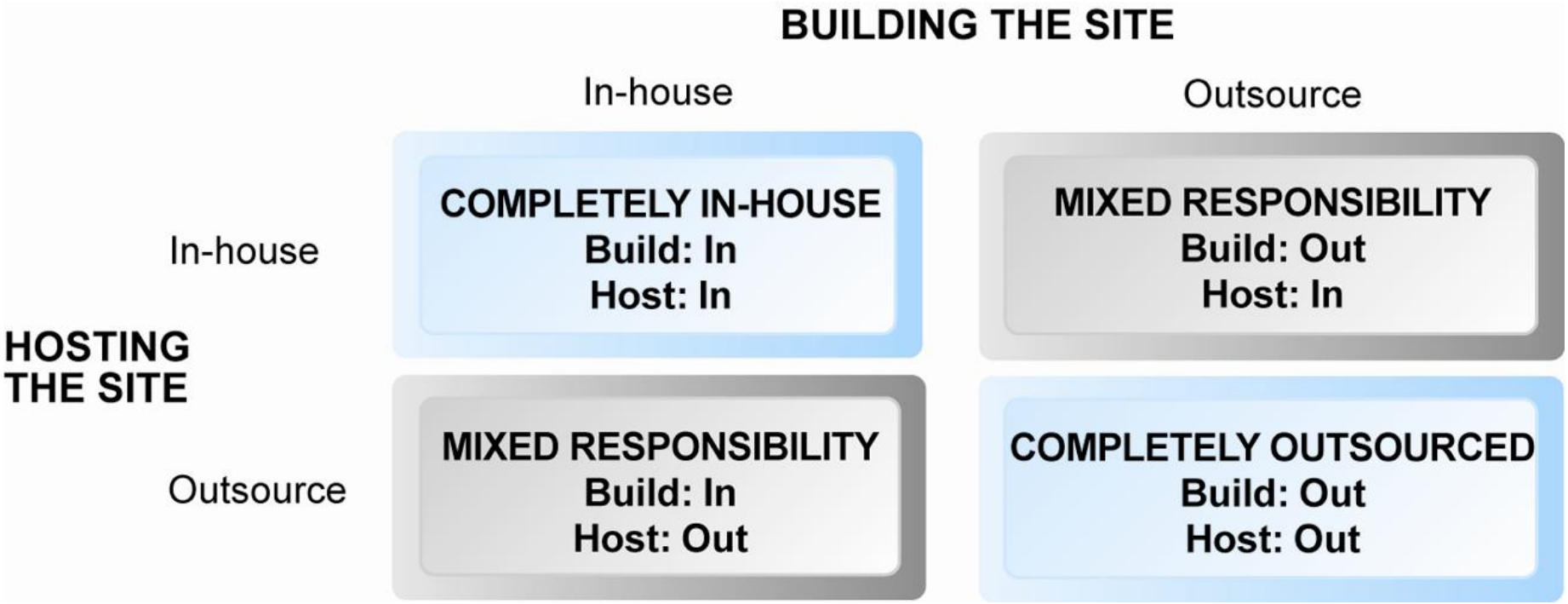


Figure 4.7 Page 198

Testing, Implementation, and Maintenance

- **Testing**
  - Unit testing-involves testing the sites program modules one at a time.
  - System testing-Involves testing the site as a whole in the way the typical user will use the site.
  - Acceptance testing-Verifies that the business objectives of the system as originally conceived are in fact working.
- **Implementation and maintenance:**
  - Maintenance is ongoing
  - Maintenance costs: Similar to development costs
  - Benchmarking-it is a process in which the site is compared with those of the competitors in terms of response speed , quality of layout and design.

Implementation and maintenance

- **The expenditure on sites annual maintenance may be equivalent to the cost of building it since e-commerce sites are always in a process of change improvement and correction.**
- **The tasks of a web site team at this stage is to:**
  - listen to customers feedback on the site
  - Develop systematic monitoring and testing plan
  - Benchmarking

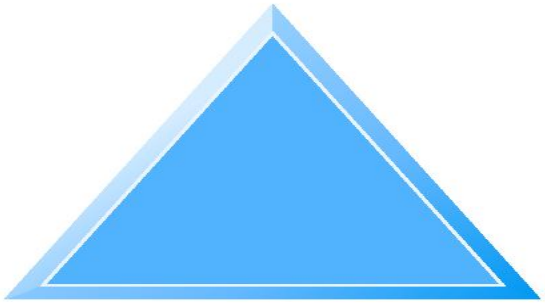
Website performance optimization

- Content and page design is under the control of the designer.
- The purpose of the website is to deliver content to customers and complete transactions. The faster this is achieved the more effective the website is from a customers perspective.
- This can be achieved by optimizing the website.

Factors in Web Site Optimization

Page Delivery

Content delivery networks
Edge caching
Bandwidth



Page Generation

Server response time
Device-based accelerators
Efficient resource allocation
Resource utilization thresholds
Monitoring site performance

Page Content

Optimize HTML
Optimize images
Site architecture
Efficient page style

Page generation-by segregating servers to perform dedicated functions and using various devices to speed up these servers

Page delivery-use of edge-caching severs e.g Akamai.

Akamai is the leading content delivery network (CDN) services provider for media and software delivery, and cloud security solutions

Edge caching is the ability to distribute content from local web servers to caching servers that are closer to the end -users

Figure 4.10, Page 205

Factors in Web Site Optimization

Page delivery can also be enhanced by using specialized content delivery networks e.g. RealNetworks or by increasing local bandwidth.

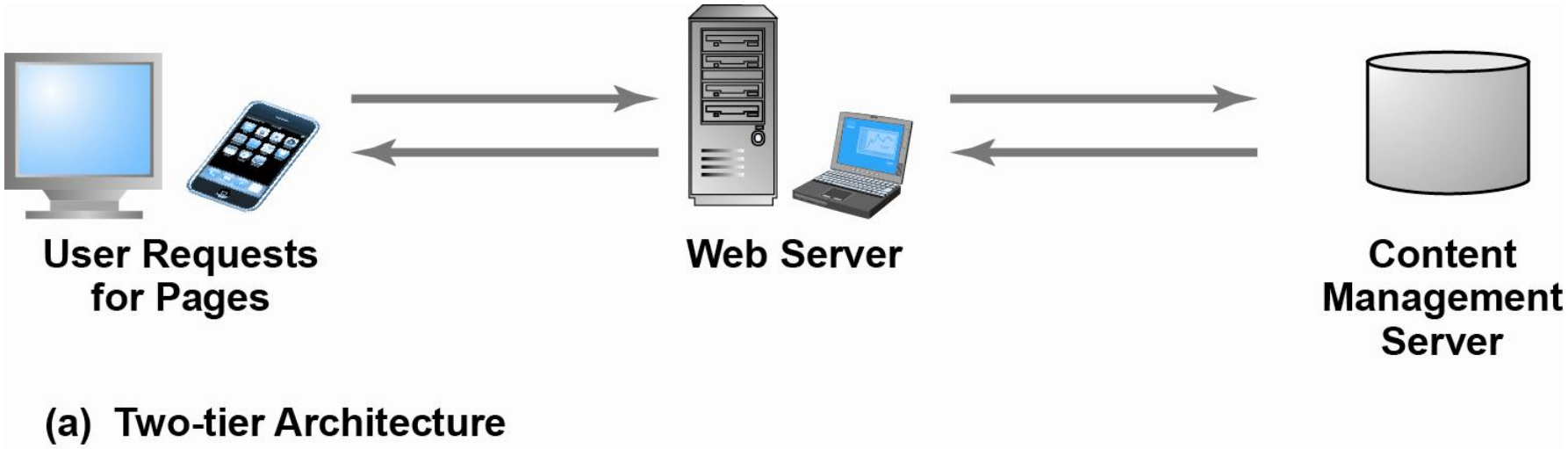
Realnetworks - a provider of Internet streaming media delivery software and services based in Seattle, Washington, United States. The company also provides subscription-based online entertainment services and mobile entertainment and messaging services.

- **Page content-** use of efficient styles and techniques for page design and content in order to reduce response time e.g reducing unnecessary HTML comments and white spaces, use of more efficient graphics, avoiding unnecessary links to other pages in the site.

Simple vs. Multi-tiered Web Site Architecture

- **System architecture**
  - Arrangement of software, machinery, and tasks in an information system needed to achieve a specific functionality
- **Two-tier**
  - Web server and database server
- **Multi-tier**
  - Web application servers
  - Backend, legacy databases

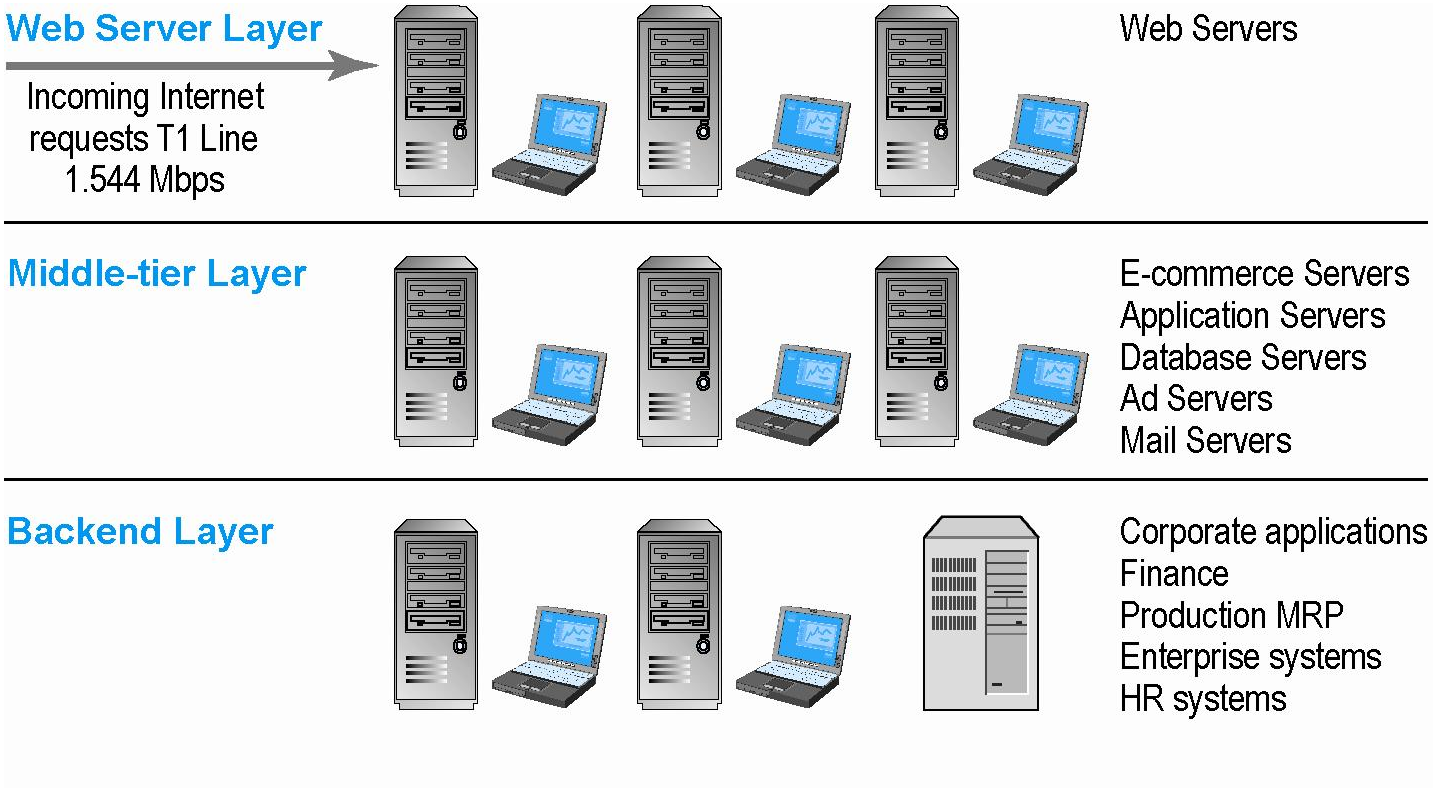
Two-Tier E-commerce Architecture



(a) Two-tier Architecture

Figure 4.11(a), Page 207

Multi-Tier E-commerce Architecture



(b) Multi-tier Architecture

In a multi-tier architecture, a Web server is linked to a middle-tier layer that typically includes a series of application servers that perform specific tasks, as well as to a backend layer of existing corporate systems.

Figure 4.11(b), Page 207

Multi-Tier E-commerce Architecture

- The development of e-commerce site requires a great deal of interactive functionality e. ability to respond to users input, take customers orders, clear credit card transactions
- This functionality requires a multi-tired systems architecture

Web Server Software

- A server software is a software used by web developers and allows them to add more webpages to their websites with ease, allows updating of webpages directly onto a website.
- Categories of webserver software includes
  - Webservers
  - Applications servers
  - Database servers
  - Cloud computing servers
  - File server software's
  - Webserver software is used to deliver content and can be accessed over the internet.

Web Server Software

- Common types of webserver sofware includes
- **Apache**
  - Leading Web server software (52% of market)
  - Works with UNIX, Linux operating systems
- **Microsoft's Internet Information Server (IIS)**
  - Second major Web server software (20% of market)
  - Windows-based (os) and compatible with a wide selection of Microsoft utility and support programs

Web Server Software

- Basic World Wide Web servers
- Apache server
- The Apache project has produced a full-featured, general purpose HTTP server, including non-forking server pool operation, an API for server extensions, content negotiation, extended logfiles, etc. Includes the complete source code. Platforms: unix.
- CERN server
- The World Wide Web daemon program, full featured, with access authorization and research tools. This daemon is also used as a basis for many other types of server and gateways. Platforms: unix, VMS.
- NCSA server
- A server for files, written in C, public domain. Many features as CERN's httpd. Platforms: unix.
- Spinner
- Spinner is a modularised, object oriented World Wide Web HTTP/1.0 and HTTP/0.9 compliant server, distributed under the GPL license.
- Alibaba
- A server for Windows NT provided by Computer Software Manufaktur.
- CL-HTTP
- CL-HTTP is a full-featured, object-oriented HTTP server written in Common Lisp by John Mallery at the M.I.T. Artificial Intelligence Laboratory. It runs on MAC, UNIX, and the Lisp Machine, and ports are underway to the PC.
- GoServe
- A server for OS/2 supporting both HTTP and Gopher, from Mike Cowlishaw of IBM UK Laboratories.
- GN
- A single server providing both HTTP and Gopher access to the same data. In C, General Public License. Designed to help servers transition from gopher to WWW. Platforms: unix.
- GWHIS Server
- Specialized WWW Servers. Commercial, from Quadralay, Inc.

Web Server Software

- **Basic World Wide Web servers**
  - HTTPS for Windows/NT
  - Service for NT. Configurable using control panel.
  - Jungle
  - Server in Tk/TCL being written by Lindsay Marshall
  - MacHTTP
  - Server for the Macintosh.
  - Netsite
  - Commercially supported server from Netscape Communications Inc.
  - Perl server
  - from Marc VanHeyningen at Indiana University. Written in perl. Platforms: unix
  - Phttpd server

    A general WWW server written in C, using features like multithreading, memory mapping and dynamic linking to achieve its goals of high speed, scalability and light weight. Written by Peter Eriksson of Signum Support AB. Platform: SunOS 5.4 (Unix)
  - Plexus
  - Tony Sander's server originally based on Marc VH's, but incorporating lots more stuff, including an Archie gateway, etc etc. Platforms: unix.
  - Purveyor
  - Process Software's server for Windows NT and future Win32 platforms such as Windows 95.
  - REXX for VM
  - A server consisting of a amall C program which passes control to a server written in REXX.
  - HTTP for VM
  - By R.M.Troth. See list of sites using it.
  - etc

Web Server Software

- Others include:
  - Google
  - Litespeed

TABLE 4.4 BASIC FUNCTIONALITY PROVIDED BY WEB SERVERS

FUNCTIONALITY	DESCRIPTION
Processing of HTTP requests	Receive and respond to client requests for HTML pages
Security services (Secure Sockets Layer)/ Transport Layer Security	Verify username and password; process certificates and private/public key information required for credit card processing and other secure information
File Transfer Protocol	Permits transfer of very large files from server to server
Search engine	Indexing of site content; keyword search capability
Data capture	Log file of all visits, time, duration, and referral source
E-mail	Ability to send, receive, and store e-mail messages
Site management tools	Calculate and display key site statistics, such as unique visitors, page requests, and origin of requests; check links on pages

Table 4.4, Page 208

Site Management Tools

- **Basic tools**
  - Included in all Web servers
    - Verify that links on pages are still valid
    - Identify orphan files:-
      - A *file* that no longer has a purpose. For example, when you uninstall an application, a few *files* that make up the application, or were generated by the application, may remain on your hard disk. These are *orphan files* since they serve no purpose without the application
- **Third-party software for advanced management**
  - Monitor customer purchases, marketing campaign effectiveness, and so on
  - WebTrends Analytics 10, Google Analytics

Dynamic Page Generation Tools

1. Dynamic page generation:

- Contents stored in database other than being hardcoded in HTML
- When a user requests for a webpage , the content of that page are fetched from a database
  - This technique is much more efficient than working directly in html code, much easier to change the contents.

2. Common tools:

- CGI (common gateway interface),
- ASP (Active server pages),
- JSP (Java server pages),
- ODBC open database connectivity

Dynamic Page Generation Tools

3. ODBC

facilities access to data from any application. It is supported by most database suppliers e.g. oracle, Sybase and IBM

Advantages of dynamic web generation tools

- Lowers menu costs
- Permits easy online market segmentation
- Enables cost-free price discrimination
- Enables content management system (CMS)-create and manage web content

Application Servers

- **Web application servers:**
- These are software's that provide specific business functionality required by web site
- It can also be viewed as a type of middleware
  - Isolate business applications from Web servers and databases
    - Single-function applications being replaced by integrated software tools that combine all functionality needed for e-commerce site

E-commerce Merchant Server Software(ecmss)

- It is a software that Provides basic functionality for sales which includes:
  - **Online catalog**
    - List of products available on Web site
  - **Shopping cart**
    - Allows shoppers to set aside, review, edit selections, and then make purchase
  - **Credit card processing**
    - Typically works in conjunction with shopping cart
    - Verifies card and puts through credit to company's account at checkout

Merchant Server Software Packages

- **Integrated environment that includes most of functionality needed to develop a sophisticated customer-centric site.**
- They include:
  - Microsoft commerce server
  - IBM Netsphere commerce
  - Broadvision commerce
  - Intershoprnfinity
  - ArtTechnology Group (ATG)

Merchant Server Software Packages

- **Key factors in selecting a package**
  - Functionality
  - Support for different business models
  - Business process modeling tools
  - Visual site management and reporting
  - Performance and scalability
  - Connectivity to existing business systems
  - Compliance with standards
  - Global and multicultural capability
  - Local sales tax and shipping rules

Web Services and Open-Source Options

- **Options for small firms**
  - Hosted e-commerce sites
    - Offer site building tools and templates
    - Example: Yahoo's Merchant Solutions
  - Open-source merchant server software
    - Enables you to build truly custom sites
    - Requires programmer with expertise, time

The Hardware Platform

- **Hardware platform:**
  - Underlying computing equipment needed for e-commerce functionality
- **Objective:**
  - Enough platform capacity to meet peak demand without wasting money
- **Important to understand the factors that affect speed, capacity, and scalability of a site**

Right-Sizing Your Hardware Platform: The Demand Side

- **Customer demand:**
  - Most important factor affecting speed of site
- **Factors in overall demand:**
  - Number of simultaneous users in peak periods
  - Nature of customer requests (user profile)
  - Type of content (dynamic vs. static Web pages)
  - Required security
  - Number of items in inventory
  - Number of page requests
  - Speed of legacy applications

Right-Sizing Your Hardware Platform: The Supply Side

- **Scalability:**
  - Ability of site to increase in size as demand warrants
- **Ways to scale hardware:**
  - **Vertically**
    - Increase processing power of individual components
  - **Horizontally**
    - Employ multiple computers to share workload ie increasing the footprint
    - Load partitioning such that some servers can handle requests while others are dedicated to handling database applications

Right-Sizing Your Hardware Platform: The Supply Side

- **Improve processing architecture**

It is a combination of vertical scaling and horizontal scaling combined of good design decisions

- **Horizontal scaling**
  - means that you **scale** by adding more machines into your pool of resources
  - **Vertical scaling** means that you **scale** by adding more power (CPU, RAM) to an existing machine.
- Examples:
  - separate static content from dynamic content and use specialized servers in each case, increase RAM and store static content in it, move CPU intensive activities on high-end servers.

TABLE 4.8 VERTICAL AND HORIZONTAL SCALING TECHNIQUES

TECHNIQUE	APPLICATION
Use a faster computer	Deploy edge servers, presentation servers, data servers, etc.
Create a cluster of computers	Use computers in parallel to balance loads.
Use appliance servers	Use special-purpose computers optimized for their task.
Segment workload	Segment incoming work to specialized computers.
Batch requests	Combine related requests for data into groups, process as group.
Manage connections	Reduce connections between processes and computers to a minimum.
Aggregate user data	Aggregate user data from legacy applications in single data pools.
Cache	Store frequently used data in cache rather than on the disk.

Table 4.8, Page 219

TABLE 4.9 IMPROVING THE PROCESSING ARCHITECTURE OF YOUR SITE

ARCHITECTURE IMPROVEMENT	DESCRIPTION
Separate static content from dynamic content	Use specialized servers for each type of workload.
Cache static content	Increase RAM to the gigabyte range and store static content in RAM.
Cache database lookup tables	Use cache tables used to look up database records.
Consolidate business logic on dedicated servers	Put shopping cart, credit card processing, and other CPU-intensive activity on dedicated servers.
Optimize ASP code	Examine your code to ensure it is operating efficiently.
Optimize the database schema	Examine your database search times and take steps to reduce access times.

Table 4.9, Page 220

Other E-commerce Site Tools

- **Web site design: Basic business considerations**
  - Enabling customers to find and buy what they need
- **Tools for Web site optimization**
  - Search engine placement
    - Metatags, titles, content
    - Identify market niches, localize site
    - Offer expertise
    - Links
    - Search engine ads
    - Local e-commerce

TABLE 4.10 E-COMMERCE WEB SITE FEATURES THAT ANNOY CUSTOMERS

- Requiring user to view ad or Flash introduction before going to Web site content
- Pop-up and pop-under ads and windows
- Too many clicks to get to the content
- Links that don't work
- Confusing navigation; no search function
- Requirement to register and log in before viewing content or ordering
- Slow loading pages
- Content that is out of date
- Inability to use browser's Back button
- No contact information available (Web form only)
- Unnecessary splash/flash screens, animation, etc.
- Music or other audio that plays automatically
- Unprofessional design elements
- Text not easily legible due to size, color, format
- Typographical errors
- No or unclear returns policy

Table 4.10, Page 221

TABLE 4.11 THE EIGHT MOST IMPORTANT FACTORS IN SUCCESSFUL E-COMMERCE SITE DESIGN

FACTOR	DESCRIPTION
Functionality	Pages that work, load quickly, and point the customer toward your product offerings
Informational	Links that customers can easily find to discover more about you and your products
Ease of use	Simple fool-proof navigation
Redundant navigation	Alternative navigation to the same content
Ease of purchase	One or two clicks to purchase
Multi-browser functionality	Site works with the most popular browsers
Simple graphics	Avoids distracting, obnoxious graphics and sounds that the user cannot control
Legible text	Avoids backgrounds that distort text or make it illegible

Table 4.11, Page 222

Tools for Interactivity and Active Content

- CGI (Common Gateway Interface)
- ASP (Active Server Pages)/ASP.NET
- Java, JSP, and JavaScript
- ActiveX and VBScript
- ColdFusion
- Web 2.0 design elements:
  - Widgets, mashups

Personalization Tools

- **Personalization**
  - Ability to treat people based on personal qualities and prior history with site
- **Customization**
  - Ability to change the product to better fit the needs of the customer
- **Cookies**
  - Primary method to achieve personalization

The Information Policy Set

- **Privacy policy**
  - Set of public statements declaring how site will treat customers' personal information that is gathered by site
- **Accessibility rules**
  - Set of design objectives that ensure disabled users can affectively access site

Designing for Accessibility

- **Why might some merchants be reluctant to make their Web sites accessible to disabled people?**
- **How can Web sites be made more accessible?**
- **Should all Web sites be required by law to provide "equivalent alternatives" for visual and sound content?**
- **What additional accessibility problems do mobile devices pose?**

Developing a Mobile Web Site and Building Mobile Applications

- **Three types of m-commerce software**
  - Mobile Web site
    - Responsive Web design
  - Mobile Web app
  - Native app
    - A native application (native app) is an application program that has been developed for use on a particular platform or device.
      - Because native apps are written for a specific platform, they can interact with and take advantage of operating system features and other software that is typically installed on that platform.
      - Because a native app is built for a particular device and its operating system, it has the ability to use device-specific hardware and software, meaning that native apps can take advantage of the latest technology available on mobile devices such as a global positioning system (GPS) and camera. This can be construed as an advantage for native apps over Web apps or mobile cloud apps.
- **Planning and building mobile presence**
  - Use systems analysis/design to identify unique and specific business objectives

TABLE 4.13 UNIQUE FEATURES THAT MUST BE TAKEN INTO ACCOUNT WHEN DESIGNING A MOBILE WEB PRESENCE

FEATURE	IMPLICATIONS FOR MOBILE PLATFORM
Hardware	Mobile hardware is smaller, and there are more resource constraints in data storage and processing power.
Connectivity	The mobile platform is constrained by slower connection speeds than desktop Web sites.
Displays	Mobile displays are much smaller and require simplification.Some screens are not good in sunlight.
Interface	Touch-screen technology introduces new interaction routines different from the traditional mouse and keyboard. The mobile platform is not a good data entry tool but can be a good navigational tool.

Table 4.13, Page 231

Developing a Mobile Web Presence

- **Design considerations**
  - Platform constraints: Smartphone/tablet
- **Performance and cost**
  - Mobile Web site:
    - Least expensive
  - Mobile app:
    - Can utilize browser API
  - Native app:
    - Most expensive; requires more programming

Building a Mobile Presence

- **What are the key differences between user experience on a Web site and on a mobile device?**
- **Why would a mobile Web site or app from the same merchant need different content or functionality?**
- **In which cases would a merchant want to develop a mobile app over a mobile Web site?**

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